Nobel Singh

Kathmandu, Nepal • nobel2073@gmail.com • github.com/nobel-sh • linkedin.com/in/nobel-singh

Experience

Twine Labs *Rust, EVM, Event Streams, Async Pipelines, PostgreSQL* Remote
Software Engineer Jan 2025 – Jan 2026

- Built **high-throughput Rust services** for event ingestion and indexing, with historical backfills, real-time processing, and **REST APIs** for querying indexed data.
- Built **sequencer-follower verification** across independent storage systems to ensure data consistency and correctness.
- Wrote **integration tests** for ingestion, synchronization, and verification pipelines under production-like conditions.

Projects

Dataquest *Next.js, TypeScript, FastAPI, PostgreSQL, Gemini API* **[GitHub]**

- Built an **AI-assisted form builder** supporting natural-language generation, manual editing, and various field types, validating generated schemas with Pydantic before persistence.
- Implemented **form versioning**, public submissions, response analytics, and CSV exports using SQLAlchemy and PostgreSQL.
- Developed **JWT authentication** with refresh-token rotation, ownership checks, and CSRF protection.

AsteroidDB *Rust, LSM Trees, Storage Engine* **[GitHub]**

- Built an embedded key-value storage engine using an **LSM-tree architecture** with memtables, SSTables, and compaction.
- Implemented **write-ahead logging**, atomic write batches, and crash recovery, with Bloom filters and block caching to accelerate reads.
- Benchmarked **116K** sequential writes/sec, **119K** random writes/sec, and **7.48M** sequential reads/sec across 1 million keys with asynchronous writes.

Subspace *Rust, Tokio, Distributed Systems* **[GitHub]**

- Built an IPFS-inspired **peer-to-peer storage prototype** using SHA-256 content addressing, chunking, and Merkle DAGs for file reconstruction.
- Implemented **Kademlia-style distributed lookups** with XOR-distance routing, k-buckets, and iterative peer discovery.
- Developed **asynchronous TCP block exchange** using Tokio, with mDNS discovery, configurable bootstrap peers, and filesystem-backed storage.

Aloha *C++, LLVM, CMake* **[GitHub]**

- Built a **native compiler** for a statically typed language, implementing lexical analysis, parsing, symbol binding, type resolution, and lowering through a custom typed intermediate representation to LLVM IR.
- Implemented structs, enums, arrays, references, and module imports, with **compile-time checks** for mutability, function arguments, exhaustive pattern matching, and missing return paths.
- Developed an **arena-backed C runtime** for allocation, strings, and dynamically sized vectors.

Education

Kathmandu University Kavre, Nepal
B.Sc. in Computer Science Dec 2021 – Feb 2026

Skills

Programming Languages: Rust, JavaScript, Python, C++
Tools & Practices: Git, GitHub, Linux, Docker, Integration Testing, Debugging
Databases: PostgreSQL, MongoDB, RocksDB

Achievements & Activities

KU Hackfest 2024 – Technical Team Leader • **NASA Space Apps Challenge 2023** – People's Choice Award • **Gajabaar Infosecurity Mentorship** – 3-month cybersecurity fellowship